

HAMILTON RBG, ON, Canada

WMO: 712970

Lat: **43.2864N**Lon: **79.9052W**Elev: **102**StdP: **100.11**Time Zone: **-5.00 (NAE)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-14.9	-22.4	0.5	-17.1	-19.3	0.7	-13.8	7.5	-2.7	6.8	-3.5	1.8	250	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.8	23.0	30.1	22.3	28.5	21.4	24.7	29.7	23.6	28.2	22.6	26.7	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.2	27.4	22.2	17.0	26.3	21.2	16.0	25.1	75.5	29.8	71.0	28.2	67.2	26.8	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.5	4.9	-20.7	34.3	4.1	2.0	-23.6	35.7	-26.0	36.9	-28.3	38.0	-31.3	39.5		
			-21.4	26.5	3.7	1.4	-24.1	27.5	-26.3	28.3	-28.3	29.1	-31.0	30.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-4.1	-3.6	0.8	7.0	13.3	18.7	21.6	20.9	17.1	10.5	4.8	-0.7	(6)
(7)		DBStd	10.18	6.19	5.56	5.39	4.67	4.51	3.86	3.16	3.02	4.02	4.54	4.51	4.80	(7)
(8)		HDD10.0	1783	437	381	289	114	16	0	0	1	49	163	333	(8)	
(9)		HDD18.3	3731	694	614	544	341	170	42	6	69	245	404	591	(9)	
(10)		CDD10.0	1389	1	0	3	24	117	259	359	338	214	66	9	1	(10)
(11)		CDD18.3	298	0	0	0	1	13	52	107	89	33	3	0	0	(11)
(12)		CDH23.3	2630	0	0	0	16	147	484	956	724	280	23	0	0	(12)
(13)		CDH26.7	799	0	0	0	2	40	149	324	204	77	3	0	0	(13)
(14)	Wind	WSAvg	2.0	2.4	2.4	2.4	2.5	2.0	1.7	1.5	1.4	1.5	1.8	2.1	2.3	(14)
(15)	Precipitation	PrecAvg	884	71	62	64	76	80	79	82	77	77	73	73	69	(15)
(16)		PrecMax	1102	148	115	140	145	146	196	187	159	173	150	133	137	(16)
(17)		PrecMin	686	30	33	27	31	26	23	18	30	25	17	21	36	(17)
(18)		PrecStd	109	29	22	31	30	30	40	42	31	34	33	28	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	14.1	18.1	25.7	30.2	32.4	34.0	33.5	31.9	26.2	19.1	13.7	(19)
(20)		0.4%	MCWB	12.2	9.7	13.1	16.9	21.6	23.0	23.8	24.1	23.4	20.2	14.0	11.1	(20)
(21)		2%	DB	9.2	9.1	14.7	21.6	27.3	30.6	32.1	30.6	28.9	22.9	16.1	10.3	(21)
(22)		2%	MCWB	7.7	7.2	10.4	14.5	19.0	22.4	23.0	22.7	21.9	17.9	12.6	8.6	(22)
(23)		5%	DB	6.1	5.7	11.4	18.3	24.2	28.3	30.2	28.9	26.6	20.3	14.3	7.6	(23)
(24)		5%	MCWB	4.5	3.4	8.2	12.3	17.6	21.6	22.3	21.7	20.7	16.3	11.2	6.1	(24)
(25)		10%	DB	3.8	3.6	8.3	14.9	21.6	26.1	28.3	27.2	24.3	17.8	12.0	5.6	(25)
(26)		10%	MCWB	2.6	1.8	5.7	10.4	16.1	19.6	21.1	20.9	19.6	14.7	9.4	4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	9.6	14.3	18.3	22.5	25.1	26.5	25.8	24.2	21.8	15.2	11.6	(27)
(28)		0.4%	MCDWB	13.4	12.0	16.8	23.4	27.8	30.3	31.5	29.7	29.7	25.0	17.0	13.2	(28)
(29)		2%	WB	8.0	6.5	11.5	15.8	20.6	23.5	24.9	24.3	23.1	18.8	13.4	8.7	(29)
(30)		2%	MCDWB	9.0	8.5	13.5	20.1	26.0	28.9	29.8	28.9	28.1	21.6	15.9	10.2	(30)
(31)		5%	WB	4.7	3.5	8.6	13.2	18.6	22.3	23.6	23.1	21.7	16.8	11.4	6.3	(31)
(32)		5%	MCDWB	5.8	5.0	11.5	17.6	22.8	27.0	27.8	27.1	25.0	19.6	13.7	7.4	(32)
(33)		10%	WB	2.7	1.8	5.8	10.5	16.7	20.9	22.5	22.1	20.5	15.3	9.8	4.5	(33)
(34)		10%	MCDWB	3.7	3.1	8.2	14.4	20.7	24.9	26.3	25.7	23.4	17.4	11.8	5.6	(34)
(35)	Mean Daily Temperature Range		MDBR	7.4	8.1	8.5	9.6	10.6	10.6	11.0	10.9	10.9	9.3	7.9	6.4	(35)
(36)		5% DB	MCDDBR	8.7	10.5	12.6	15.6	14.3	13.4	13.0	12.7	13.6	12.6	11.3	8.6	(36)
(37)			MCWBR	7.8	8.8	9.0	10.0	8.0	7.0	6.3	6.1	7.4	8.1	8.4	7.3	(37)
(38)			MCDDBR	8.3	9.9	12.1	14.2	13.1	11.9	11.2	10.9	11.7	11.3	10.3	8.4	(38)
(39)		5% WB	MCWBR	7.9	8.6	9.5	10.4	8.3	6.9	6.3	6.0	7.0	7.8	8.2	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.295	0.337	0.337	0.383	0.406	0.443	0.435	0.420	0.386	0.365	0.313	0.297	(40)	
(41)		taud	2.392	2.237	2.386	2.294	2.282	2.203	2.249	2.279	2.354	2.396	2.508	2.447	(41)	
(42)		Ebn at Noon	839	846	903	882	868	833	836	836	839	810	816	804	(42)	
(43)		Edh at Noon	77	107	105	125	130	142	134	126	108	92	70	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.41	2.17	3.41	4.44	5.40	5.92	5.92	5.20	4.15	2.51	1.56	1.07	(44)	
(45)		RadStd	0.14	0.23	0.36	0.48	0.55	0.41	0.42	0.27	0.39	0.23	0.21	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page